EE272 Homework 8

Analog Design using Cadence

Due: March 9, 2026, 11:59PM

Released: March 2, 2026

1. PDK Setup

First, make sure that you have signed the TSMC NDA:
https://na4.documents.adobe.com/public/esignWidget?wid=CBFCIBAA3AAABLblqZhD5fqB2hvYgNEGVG5Dagkrpd9iVWJnRR_QGqwZDJfvmiWS5m6VWs8LvVmVKMS4lTY4*

This must be done before you can access the TSMC 180nm process design kit (PDK). Once it has been signed, you will receive an email confirmation. Please forward this email to the TA (siddhantgupta@stanford.edu) so that your PDK access can be enabled.

Next, navigate a browser window to the Stanford FarmShare OnDemand website: https://ondemand-01.farmshare.stanford.edu/pun/sys/dashboard and launch a "Tiny" size Caddyshack Desktop interactive session. Once it has launched, click on the "Launch Caddyshack Desktop" button to start the VNC session in a new browser tab. Right click on the desktop and select "Open Terminal Here".

In the terminal, type:

```
ssh -XY caddy.best.stanford.edu
```

This will log you into the next available Caddy node. Type the following command to make sure that you have access to the TSMC180 PDK:

```
cd /cad/tsmc180_pdk/
```

If you get an error at this point and you have already forwarded your signed NDA confirmation to the TA, please contact the TA and ask for help.

Type the following command to go back to your home directory:

```
cd ~
```

Complete the TSMC180 PDK setup by running the following command:

```
bash /cad/tsmc180_pdk/ee272_setup.sh
```

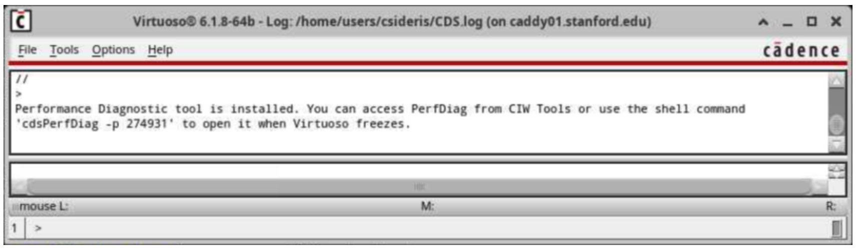
If successful, you should see the following output:

```
[csideris@caddy01 ~]$ bash /cad/tsmc180_pdk/ee272_setup.sh
Setup complete.
Copied contents of /cad/tsmc180_pdk/env_setup into /home/users/csideris/ee272_demo/env
[csideris@caddy01 ~]$
```

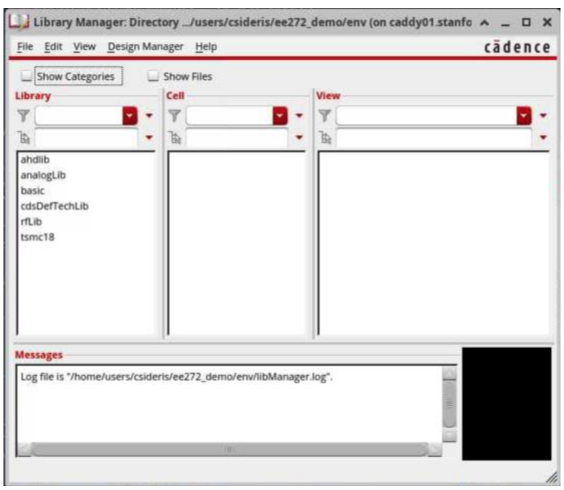
Next, type the following commands:

```
cd ee272_demo/env/
bash
source ee272_start.sh
virtuoso -64 &
```

The last line may take a little while, but it should launch the Cadence Virtuoso Command Interpreter Window (CIW), which will look something like this:



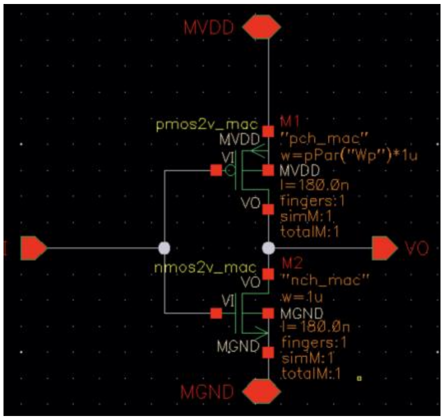
Click Tools → Library Manager. This should launch the following view:



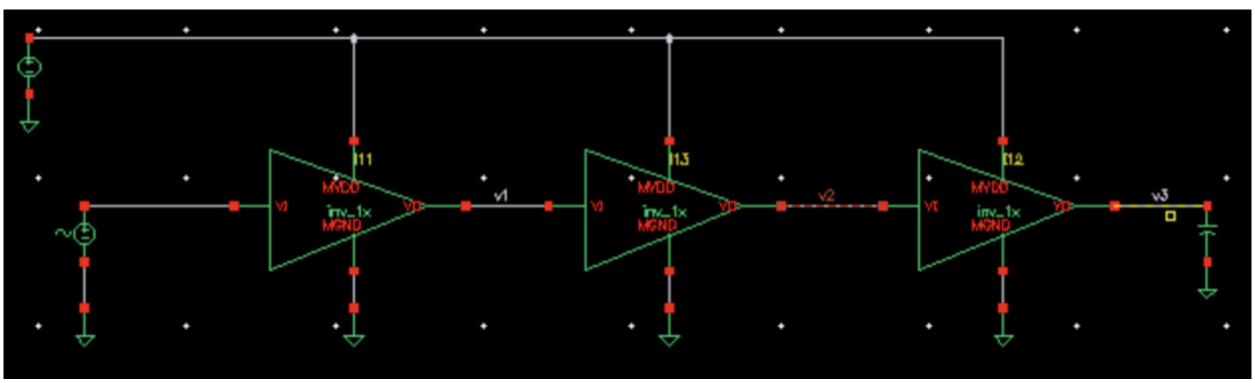
Make sure that the "tsmc18" library is listed under the Library list in the leftmost column. Now, you will need to create your own design librry. Click on File → New → Library. Under name type "ee272" and click OK. You may get an error about it failing to create the file, but if you try clicking OK again it should work at this point. At this point, you should see a dialog box pop up asking about a technology file. Select the option "Attach to an existing technology library" and click OK. Select "tsmc18" and click OK. After the process completes, you will see the TSMC 180 PDK intro splash dialog box. Click "Close".

2. Inverter Sizing

Select your "ee272" library in the Library Manager, go to File → New → Cell View. Name the cell "myinv_1x" and click OK. This will launch the Schematic L schematic editor window. Press the "i" key on your keyboard to insert components. Find and use the "nmos2v" and "pmos2v" components in the "tsmc18" library to put down one NMOS and one PMOS device in your schematic. You may find it helpful to check the "Show Categories" box. The "nmos2v" and "pmos2v" components are under the Mosfet → Mosfet_4T category. Make sure that you select the "symbol" view after which you will see parameters that you can modify for that specific instance of the device, such as the device width, length, and number of fingers. Choose 1 finger for both the NMOS and PMOS devices and choose "1u" as the finger width for the NMOS device. Since you will need to find the right sizing ratio for achieving a 50% duty cycle, you may want to use a variable to represent the PMOS size. You can achieve this by using pPar, e.g., you could choose pPar("Wp")*1u as the finger width of the PMOS device. Connect your devices using wires to form an inverter circuit. Use ports (press "p" on your keyboard to create a new port) labeled "MVDD", "MGND", "VI", and "VO" for the supply, ground, input, and output pads. Make sure to use "power" and "input/output" port types for the supply/ground ports, "signal" and "input" for VI, and "signal" and "output" for VO. Note, the pin names should be in all capital letters. Once done, your schematic should look similar to this:



Now you can create a symbol view for this inverter by going to Create → Cellview → From Cellview and choosing a Symbol view. Arrange the ports however you would like, save the symbol, and then close the window. Go to the Library Manager, and create a new schematic cell view for the test bench, which you could name "TB_inv", for example. Use the "i" key to instantiate 3 of your inverter cells, 1 "vdc" source for the supply, 1 "vsin" source to drive the input of the first inverter, and one 2fF ideal capacitor to load the output of the last inverter. You should see the "Wp" parameter for each inverter. Set all of these to another variable, e.g., "myWp". Connect the 3 inverters in a chain. Your resulting schematic should look like this:



Make sure your supply is set to 1.8V and configure your "vsin" source to produce a 1GHz sine wave from 0 to 1.8V (1.8Vpp). Save the design. Launch the simulation environment by going to Launch → ADE L. Add the "myWp" variable on the left panel in ADE L under Design Variables and set it to some number (e.g., 1) by default. Add a transient simulation (remember to check the "Conservative" box under simulation accuracy) for 10ns. Run the simulation to make sure everything is working and plot the output node of the second inverter (labeled "v2" in the screenshot above). If everything is working properly, the plot viewer should pop up, and you should see a 1GHz square wave. You can use the "dutyCycle" command in Calculator on this net to automatically compute the duty cycle of the signal. Now, you must find the optimal myWp (corresponding to the PMOS to NMOS ratio) that results in a 50% duty cycle. You could do this via trial and error manually, but it is likely easier to set up and run a parametric sweep under Tools → Parametric Analysis.

Please include a screenshot of the sinusoid input signal overlayed with the "v2" output node and report the optimal PMOS to NMOS ratio that you found.

3. Inverter Layout

Now, we are ready to layout the inverter. First, replace the pPar("Wp") parameter with the optimal Wp that you found in the previous problem. Next, create a layout view by going to Launch → Layout XL from the Schematic view of your "inv_1x" cell. Use the "i" key to insert "nmos2v" and "pmos2v" devices from the "tsmc18" PDK library. Make sure you select the "layout" view. You may need to press the keys "Shift + F" to view the layers inside the device pcells. Complete the layout of the inverter by routing using the polysilicon and metal layers. Do not forget to connect the psub to ground and the nwell to VDD. Note that this can be done automatically by choosing the "bodytie_typeL" to be "Integrated" for the NMOS and PMOS devices in their parameters. The "o" key can be used to instantiate vias. When done with the routing, make sure to add pins using the "p" key for all of the ports ("MVDD", "MGND", "VI", and "VO"). Note that the port labels must use the "pin" layers of the corresponding metal that they are on (e.g., "METAL1 pin"). You may not place ports on the polysilicon layer.

Once done with your layout, you should check the DRC for errors by going to Calibre → Run nmDRC. Set the rules file to "/cad/tsmc180_pdk/pdk/Calibre/drc/calibre.drc", choose a run directory, and click on "Run DRC". This may take a little while and should bring up the viewer once completed. Select the Filter to "Show Unresolved" to only show DRC violations. The only violations that you should see if your design is DRC clean are minimum density errors (i.e., PO.R.3, M1.R.1, M2.R.1, M3.R.R1, M4.R.R1, M5.R.R1, and UTM40K.R.R1). If you have any additional errors, you will need to debug them in your layout, correct the issues, and rerun DRC until it is clean aside from these density violations. Once you have achieved this, take a screenshot of the viewer showing the remaining density errors and include it in your write-up.

Next, you should run LVS by going to Calibre → Run nmLVS. Set the rules file to "/cad/tsmc180_pdk/pdk/Calibre/lvs/calibre.lvs", choose a run directory, and click on "Run LVS". If your LVS passes, you should see this in the log window:

```
           #      ###################       _   _
          #       #                 #       *   *
     #   #        #     CORRECT     #         |
      # #         #                 #       \___/
       #          ###################
```

Open the RVE viewer to view specific details. Once you have fixed all the LVS errors and it passes, take a screenshot of the passing RVE viewer window and include it in your writeup.

Now, we are ready to create a Calibre view with extracted parasitics. To do this, go to Calibre → Run PEX. Set the rules file to "/cad/tsmc180_pdk/pdk/Calibre/rcx/calibre.rcx" and choose a run directory. Set the Outputs → PEX Netlist → Netlist Format to "CALIBREVIEW". Click on the "Run PEX" button. Once completed, if successful, the Calibre View Setup window should pop up. Set the Cellmap File to "/cad/tsmc180_pdk/pdk/Calibre/rcx/calview.cellmap" before clicking OK. Open the Calibre view with the schematic viewer, take a screenshot of it, and include it in your write-up.

4. Ring Oscillator

Create a new schematic view called "ring_vco" in your "ee272" library. Instantiate 3 of your inverter cells and connect them in series, connecting the output of the last one back to the input of the first. Run a transient simulation using the schematic view (not the extracted Calibre view yet) and compute the frequency of oscillation. Report this in your homework write-up. Note: You may need to inject a current pulse into a node to facilitate the ring VCO to start up or set an initial voltage condition on one of the nodes.

Next, in ADE L, enable using the calibre view by going to Setup → Environment and adding "calibre" as the very first entry in the textbox under "Switch View List". It must come before "schematic", otherwise ADE L will pull the original "schematic" view instead of the "calibre" view with the extracted parasitics. Run the simulation again and compute the new oscillation frequency. Report the oscillaton frequency of the post-layout extracted inverters. What is the difference between the oscillation frequency of the idealized schematic view compared to the post-layout extracted view?

Next, make sure that you have added ports to each of the nodes of the ring oscillator (otherwise they will not be viewable from a testbench view) and produce a symbol view and layout view as you had done for the inverter. Layout the ring VCO by instantiating 3 layout views of your previously laid out inverter and connecting them using metals. Run DRC and LVS as before and include screenshots of them passing in your write-up. Finally, extract the layout of the full ring VCO.

Run the ring VCO testbench now using the calibre view of the fully extracted ring VCO. Note: This is not the same thing as what you did previously. The previous extracted simulation considered only extracted layouts for each inverter but used idealized wiring between the inverters for the oscillator. This last extracted simulation is the most realistic as it considers all of the wiring parasitics of the full VCO layout. Report the oscillation frequency as before.

In total, you should have measured and reported 3 different oscillation frequencies: 1. Schematic only, 2. Inverters post-layout extracted but ring VCO wiring schematic only, and 3. Ring VCO fully extracted.